



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2024-2025
PRE-BOARD EXAMINATION

CLASS: X
SUBJECT: ROBOTICS & AI [SET A]

FULL MARKS: 100
TIME: 2 HOURS

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first **15 minutes**.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

This paper consists of **six** printed pages.

Attempt **all** questions from Section A and **any four** questions from Section B.
The intended marks for questions or parts of questions are given in brackets[].

SECTION A
(Attempt all questions)

Question 1

[20]

Choose the correct answers to the questions from the given options.
(Do not copy the question, write the correct answers only.)

- (i) Select the field in which the following robots are used.



- (a) Medical (b) Construction (c) Space Exploration (d) Hospitality

- (ii) Identify the type of actuator:



- (a) Linear (b) Rotary (c) Hydraulic (d) Electric

- (iii) In robotics, what does the term '**degree of freedom**' refer to?
- (a) The power source of the robot.
 - (b) The number of independent movements a robot can make.
 - (c) The speed at which a robot can travel.
 - (d) The size of the robot's processor.
- (iv) What is the **primary** function of sensors in a robotic system?
- (a) To provide power to the robot.
 - (b) To process information.
 - (c) To gather data from the environment.
 - (d) To connect to the internet.
- (v) **Assertion (A):** Cobots are designed to work alongside humans.
Reason (R): Cobots are more intelligent than autonomous robots.
- (a) Both **A** and **R** are true, and **R** is the correct explanation of **A**.
 - (b) Both **A** and **R** are true, but **R** is not the correct explanation of **A**.
 - (c) **A** is true, but **R** is false.
 - (d) **A** is false, but **R** is true.
- (vi) Which component is **NOT** typically part of a robot's system?
- (a) Actuator
 - (b) Processor
 - (c) Human operator
 - (d) Sensor
- (vii) What role does AI play in smart homes?
- (a) Controlling only the lighting systems.
 - (b) Automating and optimizing household functions.
 - (c) Serving as a security alarm system.
 - (d) Reducing the number of electrical outlets needed.
- (viii) In ____ block of **4W**, we need to focus on context/situation/location of problem.
- (a) What
 - (b) Where
 - (c) Who
 - (d) Why
- (ix) **Assertion (A):** Machine learning is a subset of AI.
Reason (R): Machine learning involves programming machines with specific tasks.
- (a) Both **A** and **R** are true, and **R** is the correct explanation of **A**.
 - (b) Both **A** and **R** are true, but **R** is not the correct explanation of **A**.
 - (c) **A** is true, but **R** is false.
 - (d) **A** is false, but **R** is true.
- (x) **Assertion (A):** Python is often used in AI projects.
Reason (R): Python is a high-level language with extensive libraries.
- (a) Both **A** and **R** are true, and **R** is the correct explanation of **A**.
 - (b) Both **A** and **R** are true, but **R** is not the correct explanation of **A**.
 - (c) **A** is true, but **R** is false.
 - (d) **A** is false, but **R** is true.
- (xi) What is one of the ethical concerns related to AI?
- (a) AI's inability to be used in any industry.
 - (b) AI's potential to outperform human intelligence.

- (c) AI's contribution to unemployment.
 - (d) AI's incapacity to learn new tasks.
- (xii) Amit wants to make an AI system which can predict the salary of an employee based on his previous salaries. To train the machine he has to feed his previous salaries. The previous salary data given to the system is known as _____ and the predicted salary data is known as _____.
- (a) Testing data, training data
 - (b) Training data, testing data
 - (c) Training data, next data
 - (d) First data, next data
- (xiii) What is a challenge in cybersecurity related to AI?
- (a) AI completely eliminates hacking.
 - (b) AI increases the speed of internet connections.
 - (c) AI can be used to detect but also to perpetrate cyber-attacks.
 - (d) AI makes all computer systems invulnerable.
- (xiv) What is the primary goal of **NARS** (New Artificial Intelligence Paradigm)?
- (a) To create a robot that can pass the Turing Test
 - (b) To develop a robot that can learn and adapt to new environments
 - (c) To create a robot that can understand and reason about human language
 - (d) To develop a robot that can perform complex tasks autonomously
- (xv) Which type of sensor is used to measure distance?
- (a) Lidar
 - (b) Sonar
 - (c) Accelerometer
 - (d) Gyroscope
- (xvi) What is the purpose of a controller in a robot?
- (a) To provide power to the robot
 - (b) To sense the environment and make decisions
 - (c) To move the robot's actuators
 - (d) To communicate with other robots
- (xvii) Which of the following is an example of a deterministic computing system?
- (a) A neural network
 - (b) A genetic algorithm
 - (c) A traditional computer program
 - (d) A robotic arm
- (xviii) Human intelligence *vs.* machine intelligence debate centers on:
- (a) Autonomous decision-making power
 - (b) Capacity for creativity and emotions
 - (c) Reaction time under stressful situations
 - (d) Accuracy in data processing
- (xix) What will be the output of the following code:

```
def modify(lst):
    lst = [10, 20, 30]
    return lst
my_list = [1, 2, 3]
modify(my_list)
print(my_list)
```

(a) [10,20,30] (b) [1,2,3] (c) [10,2,3] (d) [1,20,30]

(xx) What will be the output of the following code:

```
def func(x, y=[ ]):  
    y.append(x)  
    return y  
print(func(1))  
print(func(2))  
print(func(3))
```

(a) [1], [2], [3]
(c) [1, 2, 3]

(b) [1], [1, 2], [1, 2, 3]
(d) Error due to incorrect list declaration

Question 2

Answer the following questions

- (i) Give an example of how human and machine intelligence contribute to medical diagnosis. [2]
- (ii) *A hospital has started using robotic systems to assist with surgeries. These robots are controlled by surgeons, but the robotic arms provide a level of precision that human hands cannot achieve alone. The hospital is considering using more autonomous robots to handle other healthcare tasks, such as medication distribution, patient monitoring, and routine check-ups.* [2]
What ethical considerations should the hospital address when implementing autonomous robots for patient monitoring? Mention at least **two**.
- (iii) *A company is developing autonomous vehicles for urban transportation. These vehicles use a combination of sensors and AI algorithms to navigate roads, detect obstacles, and make real-time decisions. The AI system must choose between multiple possible actions in complex situations, like deciding when to yield at a busy intersection.* [2]
If the vehicle has to decide between colliding with a minor obstacle or risking passenger safety by braking suddenly, what ethical considerations must be taken into account in programming such decision-making algorithms?
- (iv) Provide a code snippet that demonstrates **list slicing** in Python. [2]
Use the list [1, 2, 3, 4, 5] to obtain [2, 3, 4].
- (v) Differentiate between deterministic and probabilistic computing, and explain **why** probabilistic computing is advantageous for AI systems used in weather prediction. [2]
- (vi) Explain the concept of degree of freedom (**DoF**) in a robotic arm. Why is it important to control **DoF** in robots designed for precise tasks such as surgical operations? [2]
- (vii) Discuss how **bio-inspired** robots could be advantageous in search-and-rescue operations. Provide an example of a bio-inspired robot design suitable for navigating complex terrains. [2]

- (viii) Write the output of the following python code: [2]

```
a=[1,2,3,4,5,6,7,8,9]
b=a.pop(5)
c=a.pop()
d=a.pop(-1)
print(b)
print(c)
print(d)
print(a)
```

- (ix) How does the *Turing Test* contribute to understanding the relationship between human and machine intelligence? [2]
- (x) How does automated decision-making differ from autonomous decision-making in AI? [2]

SECTION B

(Answer *any four* questions from this section)

The answers in this section should consist of the programs in Python environment or any program environment with Python as base.

Each program should be written using **variable description table** codes so that the logic of the program is clearly depicted.
Flowcharts and algorithms are not required.

Question 3

- (i) How does the use of gears affect torque and speed in a robot? Provide an example. [3]
- (ii) Discuss the role of data in training AI models. [3]
- (iii) Write a program in Python to accept the age, gender ('M','F') and number of days worked from user. Calculate and display the total salary accordingly following the instructions given in the table: [9]

Age	Gender	Wages per day
>=18 and <30	M	700
	F	750
>=30 and <=45	M	800
	F	850

If age doesn't fall into any of the above criteria then it should display a message "*Inappropriate age entered*".

Question 4

- (i) What are sensors in robotics, and how do they contribute to a robot's functionality? [3]
- (ii) What role does data integrity play in AI-based predictive systems? Give an example of a potential risk if data integrity is compromised in such systems. [3]

- (iii) Write a Python program to check if a given number is a **Cyclo number** or an **Abundant number**. [9]
A number is a Cyclo number if its first and last digit are same.
For example : 52145 is a Cyclo number.
We call a positive integer '**n**' abundant if the sum of its proper divisors is greater than itself.
For example : 18 is abundant because the sum of its proper divisors (1+2+3+6+9) is greater than itself.

Question 5

- (i) Explain how the concept of "**robots as cobots**" differs from traditional robots. [3]
- (ii) Discuss how a cyberattack on a smart mobility system could disrupt **urban life**. What preventive measures should be in place to secure such systems? [3]
- (iii) Write a function to count the number of unique characters (**ignoring spaces**) in a string. [9]
Sample Input: "Hello World!"
Sample Output: 10 (H, e, l, o, W, r, d, !)

Question 6

- (i) Differentiate between automated and autonomous robots. Why might autonomy be preferred in space exploration robots? [3]
- (ii) What is bias in machine learning? How does it leads to errors? [3]
- (iii) Write a function to find the most frequently occurring element in a list. [9]
Sample Input: [1, 2, 2, 3, 3, 3, 4]
Sample Output: 3

Question 7

- (i) *A healthcare company is using AI systems to help diagnose diseases by analyzing patient data. These systems use machine learning algorithms trained on thousands of medical records to detect patterns indicative of various conditions. However, since this data is sensitive, the company has implemented cybersecurity measures to protect it from breaches.* [3]
List **two** cybersecurity threats that the company must protect against in this system and explain their potential impact.
- (ii) Analyze the **impact** of robotics on smart home technologies. [3]
- (iii) Write a Python program to read a string and check whether it is a palindrome or not. [9]